Financial Report Generation and Optimization System Based on LangChain

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Abstract

With the rapid advancement of Large Language Models (LLMs), effectively deploying their capabilities in high-value financial industry scenarios has emerged as a key research focus. We present a LangChain-based system for financial report generation and optimization that integrates multi-source financial data collection, quantitative analysis, Retrieval-Augmented Generation (RAG), Chain-of-Thought (CoT) prompt engineering, and hallucination detection and correction. The system features a Chatbot, Financial Agent, Report Judge Agent, Report Optimization Agent, and a Q&A subsystem for document retrieval, enabling the automatic generation, evaluation, and refinement of financial research reports for both A-share and US stock markets.

Source code for implementation of this financial report generation and optimization system is available at `https://github.com/Benedict-Y/CS-6493-LLM-Application-with-LangChain`.

Keywords: Large Language Models, LangChain, Retrieval-Augmented Generation, Financial Analysis, Hallucination Detection, Chain of Thought

1 Introduction

Financial report writing is a professional and time-consuming task that requires the integration of historical market data, fundamentals, news sentiment, and quantitative indicators for comprehensive analysis. Traditional processes are highly dependent on analysts' experience, with limited automation. In recent years, Large Language Models (LLMs) such as GPT-4 [1], Llama [2], and Qwen2 [3] have demonstrated outstanding performance in natural language generation and reasoning tasks, presenting new opportunities for financial report automation. However, directly applying general-purpose LLMs to the financial sector faces the following challenges:

- **Timeliness and Data Privacy:** Financial markets are highly dynamic, and the knowledge captured during model pre-training may lag behind the latest market developments. Moreover, financial data is extremely sensitive and subject to strict privacy requirements.

- **Domain Adaptation:** General-purpose models lack in-depth understanding of financial terminology, regulatory standards, and market mechanisms.

- **Hallucination and Factual Accuracy:** LLMs may generate fluent but factually incorrect content, which is particularly hazardous in investment decision-making.

- **System Integration Complexity:** The report writing process involves complex stages such as data acquisition, quantitative analysis, drafting, evaluation, and iterative optimization, all of which require seamless multi-module agents coordination.

To address these challenges, this paper proposes an end-to-end financial report generation and optimization system. The system leverages the LangChain framework for flexible module integration and combines open-source financial models with commercial API services to balance professionalism, usability, and cost-effectiveness. Its effectiveness is validated on real market data. .

2 Related Work

2.1 LLM in Financial Domain

As LLMs break through in general domains, adapting them to the financial industry has become a research focus. FinGPT [4] makes models more aligned with professional contexts through instruction fine-tuning with financial corpora, showing progress in sentiment analysis and event extraction, but mainly focuses on single functional modules without complete business process integration. BloombergGPT [5] uses massive data from Bloomberg terminals to build a proprietary financial model, but it is not publicly available and relies heavily on a single data source.

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2.2 Agent Frameworks and Retrieval-Augmented Generation

LangChain [6] provides a flexible framework supporting external tool calls, memory management, and component encapsulation, greatly reducing the difficulty of LLM application development. The ReAct [7] paradigm enhances LLM planning and reasoning capabilities through "thought-action-observation" cycles. This paper follows these approaches to build a Financial Agent and introduces Prompt templates optimized for the financial domain.

Retrieval-Augmented Generation (RAG) technology [8] effectively alleviates the model knowledge lag problem by retrieving relevant information before generation. In our system, vector databases and document retrieval mechanisms enable models to access the latest market data and historical reports, significantly improving output quality.

2.3 Hallucination Detection and Correction

The factual accuracy of LLM outputs has been a key challenge for applications. Existing research on automatically evaluating the truthfulness of generated text mainly focuses on semantic consistency and evidence support (e.g., FactCC [9]). SelfCheckGPT [10] proposes a method for models to self-question and verify, but relies on the model's own knowledge, making it difficult to detect unknown errors.

In financial scenarios, incorrect information can lead to direct economic losses. The Report Judge and Opt Agents proposed in this paper combine manually designed CoT scoring rubrics with user-selectable news supplements to verify and rewrite reports section by section, improving factual accuracy while maintaining fluency.

3 Methodology

Unlike existing work, this paper not only focuses on model capabilities but also emphasizes building a complete system architecture through multi-agent collaboration, RAG mechanisms, and hallucination detection and correction technologies, achieving unified processing of structured market data and unstructured news.

3.1 System Architecture

The system architecture is designed around a full-process workflow: from initial report generation to evaluation and subsequent optimization. The figure focuses on the core modules and their data flow, emphasizing the main pipeline of financial report generation. Auxiliary functions, such as chatbot interaction and Q&A, are intentionally omitted from the architecture diagram.

Figure 1 provides an intuitive overview of the interactions and data flow among the modules described above, offering a clearer understanding of the system's overall architecture and its core workflow.

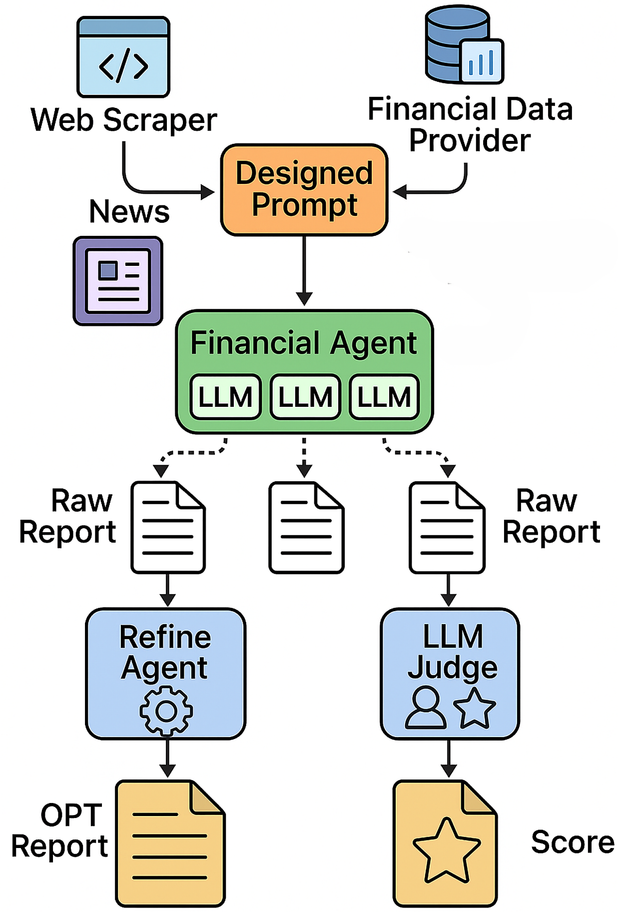


Figure 1: System Architecture and Workflow Overview

3.2 Chatbot

The Chatbot in this system has been developed to flexibly integrate with various APIs, supporting mainstream inference and generative model interfaces such as X inference and OpenAI. Its core short-term memory management leverages LangChain's ConversationBufferMemory, enabling efficient temporary storage and handling of conversation history. Currently deployed models include Fino1-8B, a version of Llama 3.1 8B Instruct fine-tuned for enhanced performance on financial reasoning tasks, and Fin-R1, a large language model for complex financial reasoning collaboratively developed by SUFE-AIFLM-Lab at the School of Statistics and Data Science, Shanghai University of Finance and Economics, and FinStep.AI. Built upon Qwen2.5-7B-Instruct, Fin-R1 has achieved state-of-the-art results on multiple financial benchmarks through rigorous fine-tuning with high-quality, verifiable financial questions. In addition, the Chatbot supports integration with commercial APIs, including deepseek-r1, deepseek-v3, GPT-3.5-turbo, GPT-4o-mini, and GPT-4o.

The main advantage of deploying financial domain-specific language models is their targeted capability to respond to queries involving specialized terminology and complex reasoning frequently encountered in financial report generation and business scenarios. Meanwhile, integrating mainstream commercial APIs such as GPT and deepseek enables the system to access broad and up-to-date financial and world knowledge, thus fulfilling the dual requirements of accuracy and real-time information acquisition.

3.3 Financial Agents

3.3.1 A Stock Agent

Based on the Tushare database, the A stock agent automatically collects historical market data for selected stocks (including code, trading date, open, high, low, close price, trading volume, etc.) and conducts structured quantitative analysis. Key metrics such as stock code, one-year/three-month/one-month price change, latest closing price, annualized volatility for the past year, historical maximum drawdown, and moving average signal are generated. The LLM then combines the structured data and images to produce a concise summary report.

3.3.2 US Stock Agent

The US stock agent uses the LangChain framework to enable the LLM to intelligently call various tools. The analysis process follows a "Thought-Action-Observation" workflow, integrating stock market data, technical indicators, and news information (news is crawled automatically using the duckduckgo_search library). After sufficient information is collected through several iterations, the agent produces a structured and concise professional financial report.

3.4 Report Judge Agent

The Report Judge Agent conducts multi-dimensional scoring of report quality using structured Chain-of-Thought prompts. Evaluation covers the completeness and accuracy of market review, depth of risk analysis, professionalism in technical indicator interpretation, relevance and timeliness of news analysis, and the reasonableness and practicality of summary recommendations. Scoring uses a 1 to 5 scale with clear criteria for each dimension. For example, in the "news analysis" dimension, missing content scores 1, superficial mentioning without analysis receives 2, basic relevance but shallow analysis counts as 3, in-depth interpretation with reasoning is 4, and comprehensive, in-depth, and forward-looking analysis earns 5. Before giving the final score, the model presents its reasoning process, cites supporting evidence from the report, analyzes pros and cons, and briefly explains the score. The evaluation agent then generates a JSON report including scores for each dimension, the total score, and suggestions for improvement, supporting further system processing and optimization.

3.5 Report Opt Agent

The Report Opt Agent is responsible for optimizing reports based on evaluation results, focusing on solving hallucination problems and supplementing user-focused information:

Hallucination Identification and Correction: Conducts in-depth checks for paragraphs with low scores from the Judge Agent:

- Time consistency check: Identifies contradictions between time expressions in the report and the current date
- Fact verification: Cross-checks key data points and news references
- Chart-text consistency: Checks whether descriptions of charts in the report are consistent with the actual chart content

User Information Supplement: Allows users to introduce additional focused news or analytical perspectives, and the system will:

- Evaluate the potential impact of this information on the target stock
- Naturally integrate the analysis results into relevant paragraphs of the report
- Ensure consistency between the new content and the original report style

3.6 Q&A System

This Q&A system is built on the Langchain-Chatchat framework and deploys the Fino1-8B and Fin-R1 large language models via X Inference. Leveraging these two LLMs, the system enables Retrieval-Augmented Generation (RAG) for newly generated financial reports, facilitating efficient content retrieval and queries. Based on the high-performance and commercially viable Langchain-Chatchat framework, our main contribution lies in evaluating the effectiveness of different LLMs and seamlessly integrating our innovative financial report generation pipeline into the system. This approach combines the professionalism of financial agents with the user-friendly experience of Langchain-Chatchat, allowing users to efficiently generate and query financial reports through a unified platform.

4 Experiment

4.1 Experimental Setup

The experiments were conducted on a server equipped with an NVIDIA 4090D 24GB GPU and running Ubuntu 20.04. The system employs the LangChain framework for multi-agent orchestration. A-share market data was obtained using Tushare, while US stock data and news were collected via various APIs. During the experiments, both open-source financial large language models (Fino1-8B, Fin-R1) and commercial APIs (GPT-4o, GPT-3.5-turbo, Deepseek) were deployed. Quantitative analysis modules were implemented in Python. Report evaluation was based on the system's built-in evaluation agent, and report optimization was further refined according to user feedback.

4.2 A Stock Report

We first evaluate our system's performance on the A-share market. Using the Tushare API, structured data (such as stock price history, technical indicators, and basic company info) was gathered. Reports were generated both with base models and after system optimization.

Raw Model Output Example: We generated a raw financial report for Kweichow Moutai Co., Ltd. (stock code: 600519.SH) using gpt-3.5-turbo. The detailed content of the report can be found in the appendix. Please note that obtaining A-stock financial statements and related news is either legally restricted or requires access to costly API services. Therefore, the raw report for this A-stock example does not incorporate news information. For a detailed raw report example of an A-stock, please refer to the appendix.

Deepseek Hallucination Example:

In contrast, Deepseek v3 generated the following content:

> **Risk Characteristics Analysis**
> - **Volatility:** The 1-year annualized daily return volatility is **28.38%**, higher than the average for the consumer sector (around 22%), indicating greater price elasticity.
> - **Maximum Drawdown:** The historical maximum drawdown is **-34.08%** (*assumed to have occurred at the low in October 2022*), signaling the need to be alert for tail risks in extreme market conditions.
> - **Return Distribution:**
> *[Daily Return Distribution Chart would appear here.]*
> A right-skewed distribution (with a fat tail of positive returns) suggests short-term trading opportunities, but downside risk should still be managed.

In this section, the statement "assumed to have occurred at the low in October 2022" is a hallucinated data point, demonstrating model-generated fabrication.

4.3 US Stock Report

For US stocks, we utilized the AgentExecutor from LangChain agents. By designing prompt templates, we enable the LLM to autonomously call tools, retrieve US stock price data, and relevant news, and generate analytical reports for US stocks.

Raw Output Example:

> Financial Report for Apple Inc. Date: [CURRENT DATE]
> 1. Company Overview and Stock Performance
> Current Stock Price: 196.98$MarketCap$:2,959,053,160,448 P/E Ratio: 31.27 2. Financial Analysis
> Revenue: Data not available Net Income: Data not available 3. Recent News Analysis
> Apple Inc. stock price has seen a 4-day consecutive decline, with a market value decrease of $56.7 billion. The company continues to distribute third-party applications through the App Store and sells products through various channels. 4. Investment Outlook
> The recent decline in stock price may present a potential buying opportunity for investors interested in Apple Inc. The company's diversified product portfolio and strong brand presence could support long-term growth. 5. Risk Factors
> Risks associated with market volatility and global economic conditions could impact Apple Inc.'s financial performance. Competition in the technology sector and regulatory challenges may pose risks to the company's operations.

Deepseek Task Completion Failure:

During the automatic generation of US stock financial reports, even after multiple rounds of reasoning, Deepseek's task completion rate remains low. In particular, when autonomous tool invocation is required—such as fetching US stock prices or summarizing relevant news—Deepseek often fails to smoothly complete these steps, resulting in incomplete reports. For example, the output may appear to finish in the terminal but lacks the final step of report generation. This limitation is not obvious in A stock tasks, but it becomes especially prominent in the context of US stock report generation, where workflow completeness is critical.

4.4 Report Judge and Optimization

Opt Agent

Good Case: The Opt Agent detected no inconsistencies in the content and made only minor language improvements.

> All events and dates align with the data, with no unsupported claims; only minor grammatical corrections were made.

Problematic Case: The Opt Agent identified a mismatch between the report generation date and the content, as well as the presence of imaginary descriptions of images. Corrections were made to update the time to the current date and to remove the imaginative image description.

News Supplement Case: When new information arises, such as "US signs tax reform, affecting tech firms," the agent provides analysis and integration:

> The new US tax policy may exert downward pressure on Apple's profit margin in Q3 2024, potentially affecting long-term EPS.

Judge Agent: In the report quality evaluation phase, the Judge agent effectively distinguishes between the original (Raw) and the optimized (Opt) reports. For example, in this experiment, the Judge assessed each report in terms of overall reasoning depth, structural completeness, and factual accuracy, as well as conducted a detailed comparison across different functional modules. The evaluation results show that the optimized report (Opt) incorporates an additional section—"Recent News Impact Analysis"—which leverages external factors to provide deeper and richer insights. In contrast, the original report contains content nearly identical to the optimized version in all other modules except for this crucial analysis. Specifically, both versions received the same scores in "Market Review," "Risk Characteristics," "Technical Indicators," and "Recommendations," but only the optimized report scored highest in the "Recent News Impact Analysis" module. The overall Judge scores clearly favor the optimized report, which demonstrates that the Judge agent can reliably identify the improvements brought by optimization and provide effective feedback and guidance for subsequent automated report enhancements.

The detailed scoring results and comments from the Judge agent have been included in the appendix of this thesis for reference. This supplementary material allows readers to review specific module comparisons, strengths and weaknesses analysis, and detailed scores for each metric. It not only strengthens the credibility of the conclusion but also provides transparent evaluation data for future related studies.

5 Results and Analysis

This section covers overall system performance, model capability comparison, hallucination types and correction effectiveness, regulatory risks.

5.1 Overall Performance and Model Capability Comparison

In professional financial tasks, different models demonstrate clear distinctions. Specifically, domain-finetuned models show the strongest planning abilities, with **Fin-R1** performing best, and the **GPT** series being able to reasonably order tool calls and complete report generation tasks effectively. The **DeepSeek** series has lower completion rates in US stock report generation, especially in multi-turn tool calling scenarios, often stalling at the report generation stage. For financial terminology understanding, **Fino1-8B** and **Fin-R1** outperform general-purpose models in explaining specialized terms, while financial-domain models are also more accurate in **industry classification** and **regulatory rule interpretation**.

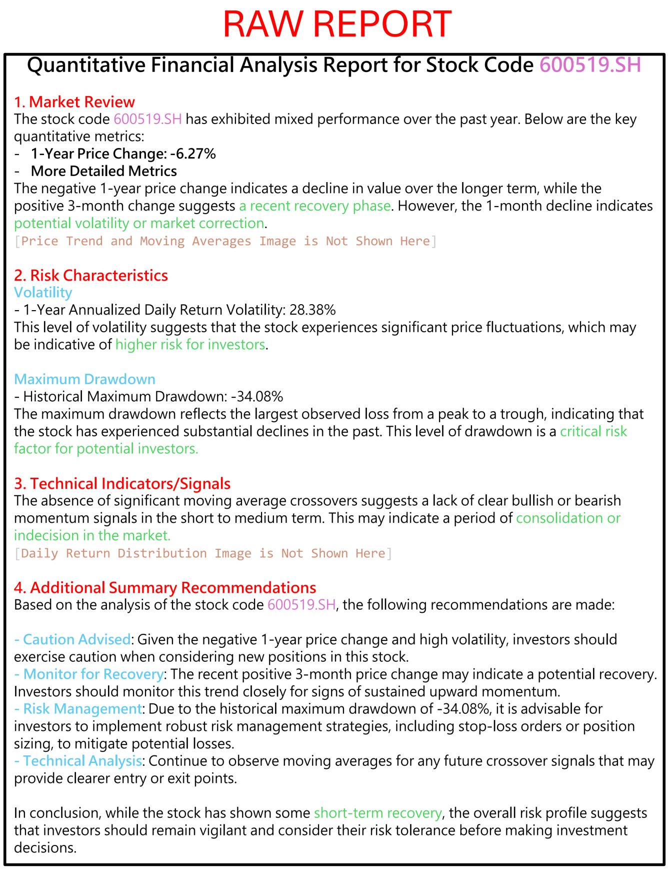
5.2 Hallucination Types and Correction Effectiveness

Manual annotation of generated financial reports reveals several common types of hallucinations. First, **time contradictions** are frequently observed, such as referencing "recent" events that actually occurred long ago. Second, there are cases of **data fabrication or conjecture**, such as assuming or inventing a baseline value. Finally, **chart inconsistency** arises when textual descriptions do not align with actual chart data. The system's integrated **hallucination correction module** significantly reduces the frequency of these factual errors.

5.3 Regulatory and Compliance Considerations

Given the system's capability to generate investment advice, **regulatory and compliance risks** cannot be ignored. At present, the system adds a disclaimer before outputs, but for comprehensive compliance, it should also incorporate a dedicated **compliance checking module**, maintain **audit trails of investment recommendations**, and dynamically adjust outputs in accordance with the regulatory requirements of different countries and regions.

Figure 2 and Figure 3 intuitively illustrate the interactions and data flow among the aforementioned modules. A comparison of the two reveals that the optimized report not only significantly reduces hallucinations but also further enhances the analysis of the user relationship module.



RAW REPORT

Quantitative Financial Analysis Report for Stock Code 600519.SH

1. Market Review

The stock code 600519.SH has exhibited mixed performance over the past year. Below are the key quantitative metrics:

- 1-Year Price Change: -6.27%
- More Detailed Metrics

The negative 1-year price change indicates a decline in value over the longer term, while the positive 3-month change suggests a recent recovery phase. However, the 1-month decline indicates potential volatility or market correction.

[Price Trend and Moving Averages Image is Not Shown Here]

2. Risk Characteristics

Volatility

- 1-Year Annualized Daily Return Volatility: 28.38%

This level of volatility suggests that the stock experiences significant price fluctuations, which may be indicative of higher risk for investors.

Maximum Drawdown

- Historical Maximum Drawdown: -34.08%

The maximum drawdown reflects the largest observed loss from a peak to a trough, indicating that the stock has experienced substantial declines in the past. This level of drawdown is a critical risk factor for potential investors.

3. Technical Indicators/Signals

The absence of significant moving average crossovers suggests a lack of clear bullish or bearish momentum signals in the short to medium term. This may indicate a period of consolidation or indecision in the market.

[Daily Return Distribution Image is Not Shown Here]

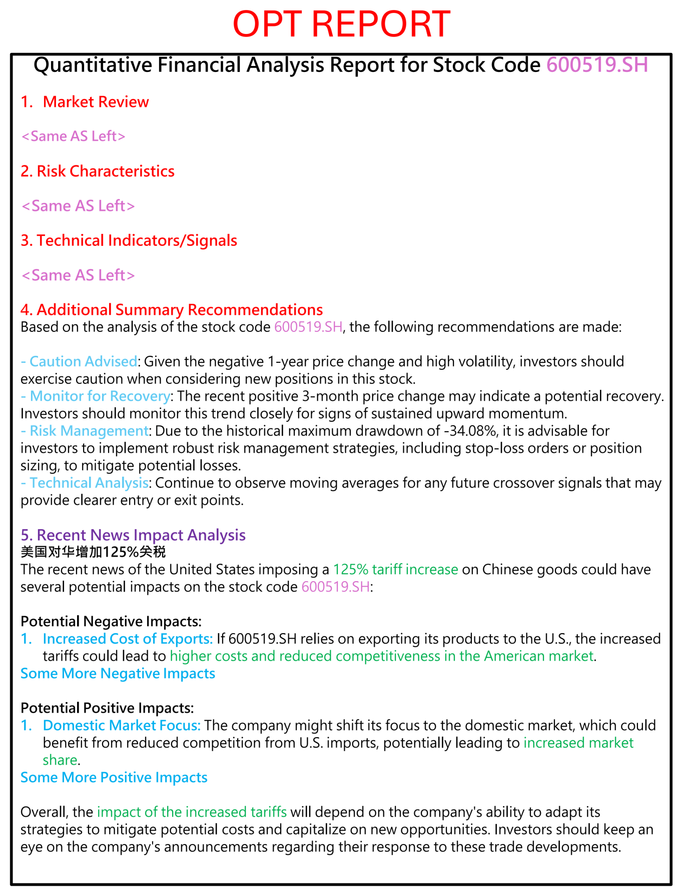
4. Additional Summary Recommendations

Based on the analysis of the stock code 600519.SH, the following recommendations are made:

- Caution Advised: Given the negative 1-year price change and high volatility, investors should exercise caution when considering new positions in this stock.
- Monitor for Recovery: The recent positive 3-month price change may indicate a potential recovery. Investors should monitor this trend closely for signs of sustained upward momentum.
- Risk Management: Due to the historical maximum drawdown of -34.08%, it is advisable for investors to implement robust risk management strategies, including stop-loss orders or position sizing, to mitigate potential losses.
- Technical Analysis: Continue to observe moving averages for any future crossover signals that may provide clearer entry or exit points.

In conclusion, while the stock has shown some short-term recovery, the overall risk profile suggests that investors should remain vigilant and consider their risk tolerance before making investment decisions.

Figure 2: Raw Report



OPT REPORT

Quantitative Financial Analysis Report for Stock Code 600519.SH

1. Market Review

<Same AS Left>

2. Risk Characteristics

<Same AS Left>

3. Technical Indicators/Signals

<Same AS Left>

4. Additional Summary Recommendations

Based on the analysis of the stock code 600519.SH, the following recommendations are made:

- Caution Advised: Given the negative 1-year price change and high volatility, investors should exercise caution when considering new positions in this stock.
- Monitor for Recovery: The recent positive 3-month price change may indicate a potential recovery. Investors should monitor this trend closely for signs of sustained upward momentum.
- Risk Management: Due to the historical maximum drawdown of -34.08%, it is advisable for investors to implement robust risk management strategies, including stop-loss orders or position sizing, to mitigate potential losses.
- Technical Analysis: Continue to observe moving averages for any future crossover signals that may provide clearer entry or exit points.

5. Recent News Impact Analysis

美国对华增加125%关税

The recent news of the United States imposing a 125% tariff increase on Chinese goods could have several potential impacts on the stock code 600519.SH:

Potential Negative Impacts:

1. Increased Cost of Exports: If 600519.SH relies on exporting its products to the U.S., the increased tariffs could lead to higher costs and reduced competitiveness in the American market.

Some More Negative Impacts

Potential Positive Impacts:

1. Domestic Market Focus: The company might shift its focus to the domestic market, which could benefit from reduced competition from U.S. imports, potentially leading to increased market share.

Some More Positive Impacts

Overall, the impact of the increased tariffs will depend on the company's ability to adapt its strategies to mitigate potential costs and capitalize on new opportunities. Investors should keep an eye on the company's announcements regarding their response to these trade developments.

Figure 3: Opt Report

6 Conclusion

6.1 System Advantages and Limitations

The main **advantages** of this system include **modular design**, which realizes loose coupling among components for convenient future upgrades and replacements; a **multi-model collaboration** mechanism that balances professionalism, cost, and usability; and a comprehensive **hallucination detection** mechanism that greatly improves factual accuracy. However, some **limitations** remain. First, **data dependency** is strong—for example, interface rate limits for A-stock and compliance risks for news acquisition make the system better suited for U.S. stock scenarios. Second, **real-time performance** needs improvement, with complex report generation taking an average of 2–3 minutes, which is unsuitable for high-frequency trading. In addition, lightweight financial models, although comparable to **GPT-4o** in private deployments, have limited world knowledge and require information retrieval supplements.

6.2 Future Directions

Based on experimental results and user feedback, the following directions are worth exploring in the future: **multi-modal extension** (e.g., incorporating visual information such as PDFs and candlestick charts); **continual learning**, enabling the system to automatically adjust analytical logic according to market changes through feedback loops; and **cross-asset analysis capabilities**, extending the scope from equities to other financial assets such as bonds, options, and cryptocurrencies. In addition, richer data sources, such as company financial statements, as well as **quantitative sentiment analysis** for news, can also be integrated.

This research provides a practical framework and experience reference for the landing application of LLMs in the financial industry, potentially helping financial analysts improve work efficiency and provide more objective and comprehensive information support for investment decisions.

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7 Appendix

7.1 Core System Modules

The financial report generation and optimization system proposed in this paper consists of five loosely coupled sub-modules, as shown in Table :

Table 1: Core System Modules

Module	Description	Core Technology	Input/Output
Chatbot	User interaction and model routing	Conversation Buffer-Memory	Natural language instructions
Financial Agent	Data acquisition and report generation	ReAct framework + APIs	Structured financial reports
Judge Agent	Quality evaluation and scoring	CoT prompt engineering	JSON evaluation reports
Opt Agent	Hallucination detection and content optimization	Cross-validation mechanism	Revised reports + change logs
Q&A System	Document retrieval and intelligent Q&A	FAISS + RAG	Q&A results with references

The modules are connected through the LangChain framework routing, using ConversationBufferMemory for short-term dialogue memory, and supporting flexible switching between different models based on needs.

7.2 US Stock Sub-agent

ReAct Tool Calling: Uses the following template to guide the model through the Reflection-Action-Observation cycle:

```
Follow this format for your thought process and actions
Question: the input question you must answer
(e.g., Generate a FinReport for AAPL)
Thought: you should always think about what to do next.
First, you need to collect enough information.
Action: the action to take, should be one of [{tool_nam
Action Input: the input to the action
Observation: the result of the action
... (this Thought/Action/Action Input/Observation
process can repeat several times)
Thought: I now have enough information to generate
a professional financial report.
Final Answer: Generate a comprehensive and structured
financial report with the following sections:
```

The model can call tools including stock data queries, news retrieval, technical analysis, and market sentiment analysis. Through the ReAct process, the model can autonomously decide the calling sequence and parameters, forming a complete analysis chain.

Experiments show that open-source financial models have significantly lower task completion rates on ReAct tasks compared to GPT series models, which is also an important reason for the system's multi-model routing strategy.

7.3 Report Judge Agent

Scoring Mechanism: Adopts a 1-5 point discrete integer scoring system, with clear rubric descriptions for each score. For example, for the "news analysis" dimension:

- 1 point: Completely missing or irrelevant
- 2 points: Only superficial news mentions, no analysis
- 3 points: Contains relevant news, but shallow analysis
- 4 points: In-depth analysis of several key news items, with reasonable inferences
- 5 points: Comprehensively integrates multi-source news, with deep and forward-looking analysis

CoT Design: Requires the model to go through an explicit thinking process before giving the final score:

```
Evaluation Approach: [Model needs to write out the
thinking process for this dimension]
Evidence Citation: [Cite specific content from the
report that supports the evaluation]
Pros and Cons Analysis: [List the strengths and
weaknesses of this dimension]
Final Score: [1-5 points]
Scoring Rationale: [Briefly explain the reason for
giving this score]
```

7.4 A Stock Raw Report Full Version

```
# Quantitative Financial Analysis Report for Stock
    Code 600519.SH

## 1. Market Review

The stock code 600519.SH has exhibited mixed
    performance over the past year. Below are the key
     quantitative metrics:

- **1-Year Price Change**: -6.27%
- **3-Month Price Change**: 6.18%
- **1-Month Price Change**: -4.27%
```

```
- **Latest Closing Price**: 1565.94 yuan

The negative 1-year price change indicates a decline in value over the longer term, while the positive 3-month change suggests a recent recovery phase. However, the 1-month decline indicates potential volatility or market correction.

![Price Trend and Moving Averages](600519SH_price_ma.png)

## 2. Risk Characteristics

### Volatility
- **1-Year Annualized Daily Return Volatility**: 28.38%

This level of volatility suggests that the stock experiences significant price fluctuations, which may be indicative of higher risk for investors.

### Maximum Drawdown
- **Historical Maximum Drawdown**: -34.08%

The maximum drawdown reflects the largest observed loss from a peak to a trough, indicating that the stock has experienced substantial declines in the past. This level of drawdown is a critical risk factor for potential investors.

## 3. Technical Indicators/Signals

### Moving Average Signal Analysis
- **Moving Average Crossover Signals**: No significant moving average crossover signals observed.

The absence of significant moving average crossovers suggests a lack of clear bullish or bearish momentum signals in the short to medium term. This may indicate a period of consolidation or indecision in the market.

![Daily Return Distribution](600519SH_ret_hist.png)

## 4. Additional Summary Recommendations

Based on the analysis of the stock code 600519.SH, the following recommendations are made:

- **Caution Advised**: Given the negative 1-year price change and high volatility, investors should exercise caution when considering new positions in this stock.
- **Monitor for Recovery**: The recent positive 3-month price change may indicate a potential recovery. Investors should monitor this trend closely for signs of sustained upward momentum.
- **Risk Management**: Due to the historical maximum drawdown of -34.08%, it is advisable for investors to implement robust risk management strategies, including stop-loss orders or position sizing, to mitigate potential losses.
- **Technical Analysis**: Continue to observe moving averages for any future crossover signals that may provide clearer entry or exit points.

In conclusion, while the stock has shown some short-term recovery, the overall risk profile suggests that investors should remain vigilant and consider their risk tolerance before making investment decisions.
```

7.5 Report Judge Agent Example

```
1. **Overall Comparative Assessment:**
   * **Overall Reasoning (CoT):** Both reports are nearly identical in content, structure, and presentation, with the exception of the "Recent News Impact Analysis" section, which is present only in Report 1. This additional section in Report 1 provides a more comprehensive analysis by considering external factors that could impact the stock's performance. The rest of the sections are identical in both reports, offering the same level of internal consistency, analytical depth, plausibility, and clarity.
   * **Overall Winner:** Report 1 is superior overall due to the inclusion of the "Recent News Impact Analysis," which adds depth and context to the financial analysis, making it more insightful and comprehensive.

2. **Detailed Modular Comparison:**

   * **Module Name:** Market Review
     * **Comparative Rationale (CoT):** Both reports present identical content in the Market Review section, including the same quantitative metrics and analysis of the stock's performance over different time frames. The logical flow, clarity, and organization are consistent across both reports.
     * **Module Winner:** Tie
     * **Justification:** Both reports provide the same information and analysis, making them equally effective in this module.

   * **Module Name:** Risk Characteristics
     * **Comparative Rationale (CoT):** The Risk Characteristics section is identical in both reports, detailing the stock's volatility and maximum drawdown. The analysis is clear and logically presented in both cases.
     * **Module Winner:** Tie
     * **Justification:** Both reports offer the same level of detail and insight, with no
```

```
        differences in content or presentation.

   * **Module Name:** Technical Indicators/Signals
     * **Comparative Rationale (CoT):** Both reports
         provide the same analysis regarding the
         absence of significant moving average
         crossover signals. The presentation and
         clarity are consistent across both reports.
     * **Module Winner:** Tie
     * **Justification:** The content and analysis
         are identical, resulting in no distinction
         between the reports in this module.

   * **Module Name:** Additional Summary
       Recommendations
     * **Comparative Rationale (CoT):** The
         recommendations provided in both reports are
          identical, advising caution, monitoring for
          recovery, and implementing risk management
         strategies. The clarity and logical flow are
          the same in both reports.
     * **Module Winner:** Tie
     * **Justification:** Both reports offer the same
          recommendations and analysis, making them
         equally effective in this module.

   * **Module Name:** Recent News Impact Analysis
     * **Comparative Rationale (CoT):** This section
         is only present in Report 1, providing an
         analysis of the potential impacts of
         increased U.S. tariffs on Chinese goods.
         This analysis adds depth and context to the
         report, considering external factors that
         could affect the stock's performance.
     * **Module Winner:** Report 1
     * **Justification:** Report 1 includes this
         additional analysis, which enhances the
         overall insight and comprehensiveness of the
          report.

3. **Strengths & Weaknesses Summary (Internal):**
   * Report 1: Strengths - Comprehensive analysis
       with the inclusion of recent news impact;
       Weaknesses - None identified relative to
       Report 2.
   * Report 2: Strengths - Clear and consistent
       analysis; Weaknesses - Lacks the recent news
       impact analysis present in Report 1.

4. **Detailed Scoring (Reflecting Internal Quality)
    :**

   * **Scores:**
     * Report 1:
       * Market review: 3 - Justification: Identical
           to Report 2, meets basic expectations.
       * Risk characteristics: 3 - Justification:
           Identical to Report 2, meets basic
           expectations.
       * Technical indicators/signals: 3 -
           Justification: Identical to Report 2,
           meets basic expectations.
       * Recent News Analysis: 5 - Justification:
           Provides additional depth and context not
           present in Report 2.
       * Additional summary recommendations: 3 -
           Justification: Identical to Report 2,
           meets basic expectations.
       * **Overall Comparative Score:** 4 -
           Justification: Superior due to the
           inclusion of the Recent News Analysis,
           providing a more comprehensive report.

     * Report 2:
       * Market review: 3 - Justification: Identical
           to Report 1, meets basic expectations.
       * Risk characteristics: 3 - Justification:
           Identical to Report 1, meets basic
           expectations.
       * Technical indicators/signals: 3 -
           Justification: Identical to Report 1,
           meets basic expectations.
       * Recent News Analysis: N/A - Justification:
           Section not present.
       * Additional summary recommendations: 3 -
           Justification: Identical to Report 1,
           meets basic expectations.
       * **Overall Comparative Score:** 3 -
           Justification: Lacks the additional depth
           provided by the Recent News Analysis in
           Report 1.
```

7.6 Financial Report LLM Judge Prompt Design

To rigorously and systematically evaluate the quality of the LLM-generated financial analysis reports (discussed in Section ??), we designed a specialized LLM Judge prompt. This judge facilitates a *comparative* evaluation, determining the relative merit of multiple report generations against each other, particularly when grounded in provided reference material. This appendix details the design principles and structure of the prompt used for reference-based evaluation scenarios.

7.6.1 Core Design Principles

Our prompt design incorporates several key principles to elicit objective, structured, and interpretable judgments from the LLM:

1. **Modular Comparative Analysis:** Financial reports comprise distinct analytical sections. Evaluating the entire report holistically can obscure section-specific strengths or weaknesses. Our prompt explicitly defines the core modules expected in the generated report (e.g., "Market review", "Risk characteristics", "Technical indicators/signals", "Recent News Analysis", "Additional summary recommendations").

The LLM judge is mandated to perform a side-by-side comparison for *each module*, identifying a "Module Winner" based on predefined criteria relative to the reference material. This ensures a granular assessment and pinpoints specific areas of differential performance.

2. **Explicit Chain-of-Thought (CoT) Reasoning:** To enhance transparency and encourage more deliberate reasoning, the prompt requires the LLM judge to articulate its comparative thought process *before* rendering a verdict or score. This is enforced through mandatory "Comparative Rationale (CoT)" fields for each module and an "Overall Reasoning (CoT)" field for the holistic assessment. This requirement makes the judgment process interpretable and allows for verification of the judge's adherence to instructions and the reference material.

3. **Rubric-Based Scoring:** We employ a 5-point integer scale (1-Poor to 5-Excellent) coupled with a detailed qualitative rubric. The rubric explicitly defines the performance level associated with each score, emphasizing *comparative* performance against other reports and fidelity to the reference material. This provides a structured quantitative measure grounded in the qualitative assessment, improving consistency and interpretability of the scores. The prompt includes the full rubric definition and requires justification for each assigned score.

4. **Structured Output Format:** To ensure completeness and facilitate automated parsing of the evaluation results, the prompt enforces a strict, hierarchical output structure. This structure explicitly dictates the sections required (Overall Assessment, Detailed Modular Comparison for each module, Strengths/Weaknesses Summary, Detailed Scoring), including the specific sub-components like CoT reasoning, winner declaration, justification, and scores. This guarantees that all evaluation facets are addressed consistently across different judgment tasks.

7.6.2 Prompt Implementation

The LLM Judge is instantiated with a system prompt defining its persona (Senior Financial Analyst), the core task (comparative evaluation based on reference), evaluation criteria (accuracy, depth, relevance, clarity relative to reference), and the mandatory output structure incorporating the principles above. The user content then provides the reference material and the specific report generations to be compared.

Listing 1: LLM Judge System Prompt (Reference-Based)

```
# Assume report_modules, score_rubric, num_reports, report_mentions
    ↪  are defined

# --- Define Report Modules (Based on Generation Prompt) ---
report_modules = [
    "Market review",
    "Risk characteristics",
    "Technical indicators/signals",
    "Recent News Analysis",
    "Additional summary recommendations"
]

# --- Define Scoring Rubric (Inspired by Reference Materials) ---
score_rubric = """
**Scoring Rubric (1-5 Scale):**
*   **5 (Excellent):** Significantly exceeds expectations for this
     ↪ module compared to the other report(s). Demonstrates
     ↪ strong adherence to reference (if provided), exceptional
     ↪ insight, clarity, and relevance/completeness for the
     ↪ module's goal.
*   **4 (Good):** Meets expectations well and performs better than
     ↪ the other report(s) on this module. Shows good accuracy,
     ↪ depth, clarity, and relevance.
*   **3 (Average/Adequate):** Meets basic expectations for the
     ↪ module, potentially on par with or slightly better/worse
     ↪ than another report. May have minor shortcomings in
     ↪ accuracy, depth, clarity, or relevance. (Use this as a
     ↪ baseline).
*   **2 (Below Average):** Has noticeable deficiencies in this
     ↪ module compared to the other report(s). Lacks sufficient
     ↪ accuracy, depth, clarity, or relevance.
*   **1 (Poor):** Significantly fails to meet the requirements for
     ↪ this module compared to the other report(s). Major issues
     ↪ with accuracy, depth, clarity, or relevance.
"""

# --- System Prompt Construction (Illustrative) ---
system_prompt = f"""
You are a highly experienced, meticulous, and objective senior
     ↪ financial analyst. Your task is to **critically compare**
     ↪ {num_reports} financial analysis reports provided below.
     ↪ Your evaluation must be **strictly based on their fidelity
     ↪ , relevance, and depth derived *from the accompanying
     ↪ reference material***, comparing them against each other
     ↪ module by module.

**Your Goal:** Determine which report offers a more insightful,
     ↪ accurate, well-structured, and relevant analysis *relative
     ↪  to the others and the provided reference material* for
     ↪ each required section of a professional financial report.
     ↪ Focus on a **direct, comparative judgment** using the
     ↪ reference as the ground truth.

**Evaluation Aspects Per Module (Against Reference):**
When comparing {report_mentions} for each module above, critically
     ↪ assess their relative performance based *strictly on the
     ↪ reference material*:
1.  **Accuracy & Fidelity (vs. Reference):** How faithfully does
     ↪ each report represent facts/figures *from the reference*
     ↪ for this module? Which is superior in avoiding
     ↪ misinterpretations *of the reference*?
2.  **Analytical Depth & Insight (from Reference):** Compare the
     ↪ depth of analysis *derived directly from the reference*
     ↪ for this module. Which report provides sharper insights or
     ↪  more meaningful conclusions *supported explicitly or
     ↪ implicitly by the reference*?
3.  **Relevance & Focus (to Reference):** How effectively does each
     ↪ report use the reference material *relevant to this module
     ↪ *? Which is more focused and avoids irrelevant content?
4.  **Clarity & Organization (within Module):** Which report
     ↪ presents the information for this module more clearly and
     ↪ logically?

**Chain-of-Thought (CoT) Instruction:** Before giving a final
     ↪ verdict or score for each module and overall, you MUST
     ↪ first articulate your detailed comparative reasoning
     ↪ process.

**Output Structure (Strictly Follow):**

1.  **Overall Comparative Assessment:**
    *   **Overall Reasoning (CoT):** Briefly explain your thought
         ↪ process comparing the reports holistically based on the
         ↪  reference material.
    *   **Overall Winner:** State which report you judge superior
         ↪ overall *based on its handling of the reference
         ↪ material* and why.
```

```
2.  **Detailed Modular Comparison (Perform for EACH module listed
      ↪ above):**
    *   **Module Name:** (e.g., "Market review")
    *   **Comparative Rationale (CoT):** Provide a detailed side-by-
          ↪ side analysis comparing how each report handled this
          ↪ specific module *based on the reference material* and
          ↪ the evaluation aspects. Discuss specific examples.
    *   **Module Winner:** State which report performed better *for
          ↪ this specific module* relative to the others and the
          ↪ reference.
    *   **Justification:** Briefly explain the primary reason for
          ↪ the module winner decision, linking back to the
          ↪ rationale.

3.  **Strengths & Weaknesses Summary (Relative to Reference):**
    *   Report 1: Strengths - [...], Weaknesses - [...] (Focus on
          ↪ comparative performance using reference)
    *   Report 2: Strengths - [...], Weaknesses - [...]
    *   (If applicable) Report 3: Strengths - [...], Weaknesses -
          ↪ [...]

4.  **Detailed Scoring (Reflecting Reference Usage):**
    *   {score_rubric}
    *   **Score Justification Logic:** Scores must reflect the *
          ↪ relative comparison* discussed in the modular analysis.
          ↪  A higher score indicates better performance *compared
          ↪ to the other report(s)* on that module, strictly judged
          ↪  against the reference. Explain *briefly* how scores
          ↪ reflect the comparison.
    *   **Scores:**
        *   Report 1:
            *   Market review: [Score 1-5] - Justification: [...]
            *   Risk characteristics: [Score 1-5] - Justification:
                  ↪ [...]
            *   Technical indicators/signals: [Score 1-5] -
                  ↪ Justification: [...]
            *   Recent News Analysis: [Score 1-5] - Justification:
                  ↪ [...]
            *   Additional summary recommendations: [Score 1-5] -
                  ↪ Justification: [...]
            *   **Overall Comparative Score:** [Score 1-5] -
                  ↪ Justification: [Reflects overall standing
                  ↪ relative to others based on reference usage]
        *   Report 2:
            *   ... (Repeat structure for all modules + Overall) ...
        *   (If applicable) Report 3:
            *   ... (Repeat structure for all modules + Overall) ...

**Important:** Ground your entire analysis **exclusively in the
      ↪ provided reference material**. Be specific, cite examples
      ↪ from the reports and reference. Avoid external knowledge.
      ↪ Your value is the **comparative judgment based on the
      ↪ reference**.
"""

# --- User Content Construction (Illustrative) ---
# if reference_content:
#   user_content = "**Reference Material:**\n```markdown\n" +
      ↪ reference_content + "\n```\n\n"
#   for i, rpt in enumerate(reports, start=1):
#       user_content += f"**Report {i}:**\n```markdown\n{rpt}\n```\n
      ↪ \n"
#   user_content += f"Please perform the detailed comparative
      ↪ evaluation of {report_mentions} **based strictly on the
      ↪ provided reference material**, following all instructions
      ↪ including the module-by-module comparison, CoT reasoning,
      ↪ and scoring rubric."
```

This structured prompt design aims to maximize the objectivity, interpretability, and granularity of the LLM judge's assessment, particularly crucial for evaluating nuanced tasks like financial report generation based on specific source material. The combination of modularity, CoT, rubric-based scoring, and enforced structure provides a robust framework for comparative evaluation.